SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F70LA1 — Life Insurance Mathematics A

Semester One- 2015-16

Duration: 2 hours.

Please answer **ALL 7** questions.

Total marks: 60

Approved electronic calculators only can be used.
You should have a copy of the yellow Formulae and Tables for Actuarial Examinations on your desk.

1. Suppose that, for integer values of x between 70 and 75 inclusive,

   $p_x = (155 - x)/100$

   Suppose also that the effective rate of interest is 3.5% per year.

   Calculate the following:

   (a) q_{71} [1 mark]

   (b) ${}_3p_{70}$ [2 marks]

   (c) The expected present value of a term assurance for a life aged 74 with a term of 2 years and a sum assured of £2 million, payable at the end of the year of death. [3 marks]

   (d) The standard deviation of the present value in (c) [3 marks]

   [Total 9 marks]

2. The function $S_0(x) = \exp\left(-\frac{x}{20} - \frac{x^3}{300{,}000}\right)$ has been proposed as a survival function.

   (a) Verify that $S_0(x)$ satisfies the three criteria for it to be a survival function. [3 marks]

   (b) Use this survival function to calculate ${}_{40}p_0$. [2 marks]

   (c) Derive an expression for the function ${}_tp_x$.
   Simplify your answer as far as possible. [3 marks]

   [Total 8 marks]

3. A select survival model has a select period of 2 years. The ultimate part is the same as ELT 15 (Females). Probabilities within the select period are given by the following formulae:

$p_{[x]} = p_{x-4}^{ELT}$

$p_{[x]+1} = p_{x-1}^{ELT}$

(a) Calculate ${}_2p_{[40]}$, ${}_3p_{[40]+1}$ and ${}_{3|4}q_{[40]}$. [5 marks]

(b) Assuming that the force of mortality between successive integer ages is constant, calculate ${}_4q_{[40]+0.8}$. [3 marks]

(c) Assuming that the force of mortality between successive integer ages is constant, calculate $\mathring{e}_{[40]}$.
[5 marks]

[Total 13 marks]

4. Calculate ${}_{2.8}q_{53.5}$ assuming a uniform distribution of deaths between successive integer ages.

The basis is PMA92C20.

[3 marks]

5. Calculate $A^{(4)}_{60:\overline{4}|}$.

You should assume the following basis, and use the claim acceleration method.

Interest: 7% per year effective

Mortality:

$q_{60} = 0.03$
$q_{60+n} = 1.2q_{59+n}$ for all n.

[5 marks]

6. On 1 January 2005, an insurance company issued special 15-year term assurance policies to a large number of lives then aged 50 exact. For each policy, the sum assured is £200,000 for the first 10 years, and £50,000 for the remaining 5 years.

   The sum assured is payable immediately on death. Level monthly premiums are payable in advance throughout the term, or until earlier death.

   The pricing basis is as follows:

   Mortality: AM92 Select

   Interest: 4% per year effective

   (a) Show that the monthly premium that is payable for each policy is approximately £63. [5 marks]

   (b) Calculate the reserve per policy as at 31 December 2014, assuming that the reserving basis is the same as the premium basis. [3 marks]

   (c) Why might the insurance company consider that this policy is designed poorly? Give two examples of how you might have improved the design if you had been responsible for its sale. [4 marks]

   (d) At 1 January 2014, there were 2000 policies still in force. There were then 9 deaths in 2014. Calculate the mortality profit or loss to the insurance company in 2014, on the basis above. Explain why there is a mortality profit. [3 marks]

   [Total 15 marks]

7. (a) Define the random variables T_x and K_x. [2 marks]

(b) State the functions of T and K for which the following are the expected values:

(i) $\overline{A}_x$

(ii) a_{20}

(iii) $A_{[30]:\overline{10}|}$

(iv) ${}_{6|}\ddot{a}_x$

[5 marks]

[Total 7 marks]

END OF PAPER